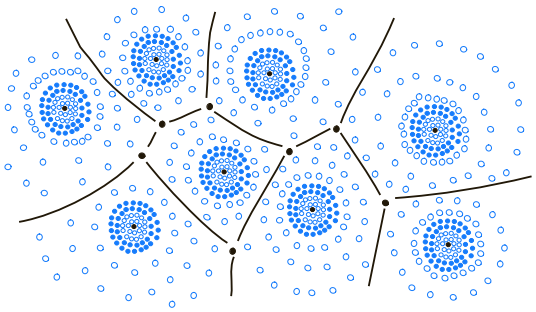
Chapter 2

Object-Oriented Design



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2.1 Goals, Principles, and Patterns

As the name implies, the main "actors" in the object-oriented paradigm are called ***objects***. Each object is an ***instance*** of a ***class***. Each class presents to the outside world a concise and consistent view of the objects that are instances of this class, without going into too much unnecessary detail or giving others access to the inner workings of the objects. The class definition typically specifies the ***data fields***, also known as ***instance variables***, that an object contains, as well as the ***methods*** (operations) that an object can execute. This view of computing fulfill several goals and incorporates design principles, which we will discuss in this chapter.

2.1.1 Object-Oriented Design Goals

Software implementations should achieve ***robustness***, ***adaptability***, and ***reusability***. (See Figure 2.1.)

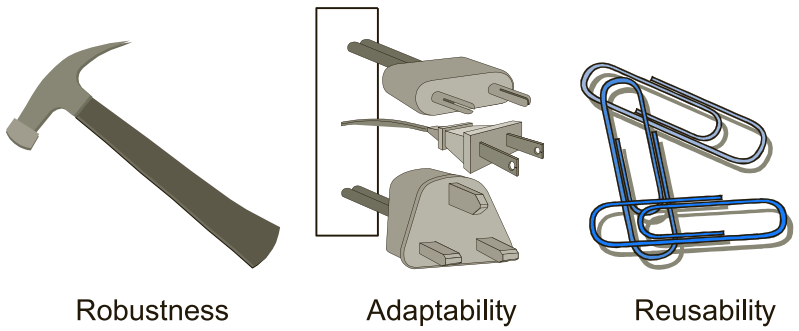

Figure 2.1: Goals of object-oriented design.

Robustness

Every good programmer wants to develop software that is correct, which means that a program produces the right output for all the anticipated inputs in the program's application. In addition, we want software to be ***robust***, that is, capable of handling unexpected inputs that are not explicitly defined for its application. For example, if a program is expecting a positive integer (perhaps representing the price of an item) and instead is given a negative integer, then the program should be able to recover gracefully from this error. More importantly, in ***life-critical applications***, where a software error can lead to injury or loss of life, software that is not robust could be deadly. This point was driven home in the late 1980s in accidents involving Therac-25, a radiation-therapy machine, which severely overdosed six patients between 1985 and 1987, some of whom died from complications resulting from their radiation overdose. All six accidents were traced to software errors.

Adaptability

Modern software applications, such as Web browsers and Internet search engines, typically involve large programs that are used for many years. Software, therefore, needs to be able to evolve over time in response to changing conditions in its environment. Thus, another important goal of quality software is that it achieves ***adaptability*** (also called ***evolvability***). Related to this concept is ***portability***, which is the ability of software to run with minimal change on different hardware and operating system platforms. An advantage of writing software in Java is the portability provided by the language itself.

Reusability

Going hand in hand with adaptability is the desire that software be reusable, that is, the same code should be usable as a component of different systems in various applications. Developing quality software can be an expensive enterprise, and its cost can be offset somewhat if the software is designed in a way that makes it easily reusable in future applications. Such reuse should be done with care, however, for one of the major sources of software errors in the Therac-25 came from inappropriate reuse of Therac-20 software (which was not object-oriented and not designed for the hardware platform used with the Therac-25).

2.1.2 Object-Oriented Design Principles

Chief among the principles of the object-oriented approach, which are intended to facilitate the goals outlined above, are the following (see Figure 2.2):

- Abstraction
- Encapsulation
- Modularity

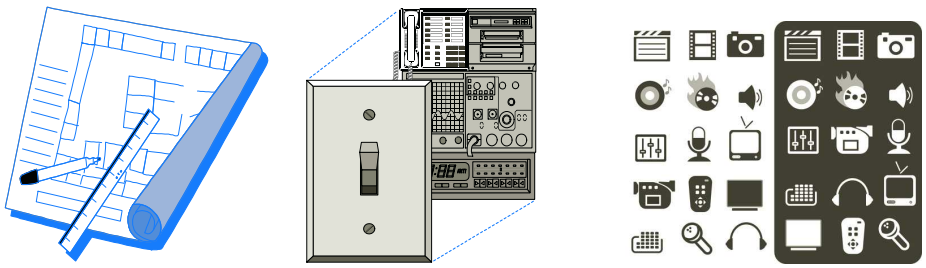

Figure 2.2: Principles of object-oriented design.

Abstraction

The notion of ***abstraction*** is to distill a complicated system down to its most fundamental parts. Typically, describing the parts of a system involves naming them and explaining their functionality. Applying the abstraction paradigm to the design of data structures gives rise to ***abstract data types*** (ADTs). An ADT is a mathematical model of a data structure that specifies the type of data stored, the operations supported on them, and the types of parameters of the operations. An ADT specifies ***what*** each operation does, but not ***how*** it does it. In Java, an ADT can be expressed by an ***interface***, which is simply a list of method declarations, where each method has an empty body. (We will say more about Java interfaces in Section 2.3.1.)

An ADT is realized by a concrete data structure, which is modeled in Java by a ***class***. A class defines the data being stored and the operations supported by the objects that are instances of the class. Also, unlike interfaces, classes specify ***how*** the operations are performed in the body of each method. A Java class is said to ***implement an interface*** if its methods include all the methods declared in the interface, thus providing a body for them. However, a class can have more methods than those of the interface.

Encapsulation

Another important principle of object-oriented design is ***encapsulation***; different components of a software system should not reveal the internal details of their respective implementations. One of the main advantages of encapsulation is that it gives one programmer freedom to implement the details of a component, without concern that other programmers will be writing code that intricately depends on those internal decisions. The only constraint on the programmer of a component is to maintain the public interface for the component, as other programmers will be writing code that depends on that interface. Encapsulation yields robustness and adaptability, for it allows the implementation details of parts of a program to change without adversely affecting other parts, thereby making it easier to fix bugs or add new functionality with relatively local changes to a component.

Modularity

Modern software systems typically consist of several different components that must interact correctly in order for the entire system to work properly. Keeping these interactions straight requires that these different components be well organized. ***Modularity*** refers to an organizing principle in which different components of a software system are divided into separate functional units. Robustness is greatly increased because it is easier to test and debug separate components before they are integrated into a larger software system.

2.1.3 Design Patterns

Object-oriented design facilitates reusable, robust, and adaptable software. Designing good code takes more than simply understanding object-oriented methodologies, however. It requires the effective use of object-oriented design techniques.

Computing researchers and practitioners have developed a variety of organizational concepts and methodologies for designing quality object-oriented software that is concise, correct, and reusable. Of special relevance to this book is the concept of a ***design pattern***, which describes a solution to a "typical" software design problem. A pattern provides a general template for a solution that can be applied in many different situations. It describes the main elements of a solution in an abstract way that can be specialized for a specific problem at hand. It consists of a name, which identifies the pattern; a context, which describes the scenarios for which this pattern can be applied; a template, which describes how the pattern is applied; and a result, which describes and analyzes what the pattern produces.

We present several design patterns in this book, and we show how they can be consistently applied to implementations of data structures and algorithms. These design patterns fall into two groups—patterns for solving algorithm design problems and patterns for solving software engineering problems. Some of the algorithm design patterns we discuss include the following:

- Recursion (Chapter 5)
- Amortization (Sections 7.2.3, 11.4.4, and 14.7.3)
- Divide-and-conquer (Section 12.1.1)
- Prune-and-search, also known as decrease-and-conquer (Section 12.5.1)
- Brute force (Section 13.2.1)
- The greedy method (Sections 13.4.2, 14.6.2, and 14.7)
- Dynamic programming (Section 13.5)

Likewise, some of the software engineering design patterns we discuss include:

- Template method (Sections 2.3.3, 10.5.1, and 11.2.1)
- Composition (Sections 2.5.2, 2.6, and 9.2.1)
- Adapter (Section 6.1.3)
- Position (Sections 7.3, 8.1.2, and 14.7.3)
- Iterator (Section 7.4)
- Factory Method (Sections 8.3.1 and 11.2.1)
- Comparator (Sections 9.2.2, 10.3, and Chapter 12)
- Locator (Section 9.5.1)

Rather than explain each of these concepts here, however, we will introduce them throughout the text as noted above. For each pattern, be it for algorithm engineering or software engineering, we explain its general use and we illustrate it with at least one concrete example.